Derbyshire Shared Care Pathology Guidelines

Hypernatraemia in Adults

Purpose of Guideline

The investigation and management of adult patients with newly diagnosed hypernatraemia.

Hypernatraemia is defined as a plasma sodium greater than 146 mmol/L.
The clinical significance of hypernatraemia depends on its severity, speed of onset and underlying cause.
It is usually due to a deficiency of water, rather than an excess of sodium.

The laboratory will telephone new sodium results 155 mmol/L or above to the GP practice or GP out of hours service/111.

Symptoms

Primarily neurological and reflect changes in brain volume (shrinkage). They range in severity and include:

- Cognitive dysfunction e.g. confusion, decreased consciousness
- Dehydration & thirst
- Headache
- Nausea and vomiting
- Lethargy
- Irritability or even seizures
- Nystagmus
- Myoclonic jerks

There is no specific concentration at which these symptoms may occur and hypernatraemia developing over days to weeks may be relatively asymptomatic, even with Na >155 mmol/L. There may be additional symptoms which reflect the underlying cause.

Even mild hypernatraemia is an extremely potent stimulus of thirst, although this diminishes over time. Therefore, it usually only develops if there is restricted access to fluids or an inability to express thirst.

Orthostatic hypotension, tachycardia. Decreased urinary output and other signs of hypovolaemia may also be present.

Risk factors for Hypernatraemia

- Age >65
- Dementia, other mental or physical disability
- Residential care settings

What are the main causes of hypernatraemia?
Hypernatraemia is almost always due to excess water loss or inability to replace normal insensible loss by drinking.

Excessive intake of sodium is an uncommon cause, but may be suggested by a non-elevated urea, and could raise the possibility of deliberate self harm or abuse. This is difficult to diagnose, but if suspected it is useful to check serial weights and paired blood and random urine samples.

There are three common causes for hypernatraemia

1. **Low fluid intake or excess fluid loss**
   Anything which impairs thirst, swallowing or access to water. Fluid losses will be exacerbated by diarrhoea, fever, high ambient temperature or thyrotoxicosis.

2. **Arginine Vasopression Deficiency (AVP-D) and Arginine Vasopressin Resistance (AVP-R) (previously known as Cranial and Nephrogenic Diabetes insipidus (DI) respectively**
   This may be central (pituitary), due to lack of AVP (also known as ADH) secretion, or nephrogenic, due to renal resistance to AVP. Polyuria and polydipsia are the main symptoms. The commonest cause is treatment with lithium which causes AVP-R. Any history of current or previous lithium use is relevant: even if levels have always been within the therapeutic range and even if lithium use was a long time ago. AVP-D may result from head injury or pituitary disease.

   Hyperosmolar hyperglycaemic state (HHS) This is a decompensated form of type 2 diabetes. Severe, prolonged hyperglycaemia causes osmotic diuresis with a net loss of free water. Plasma sodium tends to be diluted by the osmotic pull of water from the intracellular to the extracellular space, thus the “true” sodium is higher still.

Other causes
Endocrine disorders causing excess adrenal steroid secretion (i.e. Cushing’s and Conn’s syndromes) rarely cause hypernatraemia, but sodium may be at or just above the upper limit of normal (146 mmol/L). There are usually other clinical features present, e.g. hypertension, hypokalaemia, signs of Cushing’s.

What other information is needed in the hypernatraemic patient?
- Symptoms and signs of neurological disturbance
- Presence or absence of thirst
- Fluid balance information: is the patient drinking? Is there polyuria/polydipsia? Is there diarrhoea?
- Blood pressure
- Volume status: hypovolaemia, euvolaemia or oedema
- Medical and drug history

What other investigations are needed in the hypernatraemic patient?

- U&E
- Glucose
- Serum Osmolality
- Urine Na and osmolality

Urine osmolality (on a random sample) is the single most important test:

- When lack of water intake is the cause, the urine will be maximally concentrated, i.e. osmolality >750 mmol/kg
- In AVP-D and AVP-R the urine is inappropriately dilute (<750 mmol/kg), and usually lower than the plasma osmolality
- In HHS the urine and plasma osmolalities are similar to each other, i.e. around 350-450 mmol/kg and glucose is markedly raised

Which patients should be referred?

Indications for urgent admission to hospital are:

- Na >155 mmol/L
- Na 146-155 mmol/L with neurological disturbance or an inability to drink adequately
- Presence of HHS, i.e. hypernatraemia plus hyperglycaemia >30 mmol/L

Treatment

Hypernatraemia indicates that water intake is insufficient. If the cause of this cannot be rectified, admission for intravenous fluids will be necessary. If oral fluids can be increased and plasma sodium is <155 mmol/L it may be possible to manage without admission, but daily review and frequent U&E testing will be required. Excessively rapid reduction in plasma Na with intravenous hypotonic fluids may cause acute cerebral oedema.

Contacts

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Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
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A&E, MAU and ACC	Via RDH switchboard, 01332 340131(24/7)
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Duty Biochemist	01283 593313 (8am - 4pm Mon - Fri)
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